COMSATS University Islamabad, Lahore Campus

☑ **Midterm Examination** ☐ **Terminal Examination – FALL 2021**

Course Title:	Design and Analysis of Algorithms				Course Code:	CSC301	Credit Hours: 3(3,0)
Course Instructor:	Dr. Hasan Jamal				Programme Name:	BS Computer Science	
Semester:	5th	Batch:	FA19-BCS	Section:	A. B. C	Date:	17/11/2021
Time Allowed:	**90 Minutes**				**Maximum Marks:**		**60**
Student's Name:					Reg. No.		

Important Instructions / Guidelines:

- Be precise and to the point while answering any question
- Show all your work, as partial credits will be given. You will be graded not only on the correctness of your answer, but also on the clarity with which you express it. Please be neat.
- **Good luck!**

Section 1

Question 1: **[Marks: 4]**

If an algorithm provides a correct solution to a problem, why do we still want to analyze the algorithm?

Question 2: **[Marks: 10]**

Prove or disprove: $\frac{1}{4}n^2 - 8 \in \Theta(n^2)$

Question 3: **[Marks: 10]**

Below is the pseudocode for the iterative algorithm SORTEDINSERT, which inserts the key k into an already sorted array A and maintains the sorted order of the array.

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Owner of SuperCUI

```
SORTEDINSERT(A,k)
1   i = length(A)
2   while i > 0 and k < A[i]
3       A[i + 1] = A[i]
4       i = i − 1
5   A[i + 1] = k
```

a) Perform a line-by-line analysis of SORTEDINSERT and derive a precise expression of the running time T(n), where n = length(A).

b) Describe the best-case scenario for SORTEDINSERT and determine its asymptotically tight bound on the best-case running time.

c) Describe the worst-case scenario for SORTEDINSERT and determine its asymptotically tight bound on the worst-case running time.

Section 2

Question 4: **[Marks: 2.5 x 4 = 10]**

In not more than five lines each, provide the answers to the following questions. Be reasonable in your assumptions. Justify your answer.

a) You are given a list containing the birth year of all the people residing in Lahore. Your task is to determine the year in which the maximum number of people were born. In terms of time complexity, what is the best way to solve this problem?

b) You have a uniformly distributed set of unsorted integers of 10,000 numbers, within the range 18,000 to 25,000. Which algorithm will you choose to sort this data set? Justify your answer.

c) You are given an unsorted list of NTS exam marks of students applying for admission at CUI Lahore. Which algorithm will you choose to sort this data set? Justify your answer.

d) You have an unsorted list of 1,000 numbers, within the range 22,000 to 59,000. Which algorithm will you choose to sort this data set? Justify your answer.

Question 5: **[Marks: 10]**

Solve the following recurrence using the "Recursion Tree Method"

$$T(n) = 2\,T(n-1) + 1$$

Question 6: **[Marks: 4]**

Solve the following recurrence using the "Master Theorem Method"

$$T(n) = 2T\left(\frac{2n}{3}\right) + n$$

Question 7: **[Marks: 6 + 6 = 12]**

Given a hash table of size 11 and a hash function H(k) = K mod 11. Draw the contents of the hash table if:

a) *Open* addressing with linear probing is used to resolve collisions. What values will be in the hash table after the following sequence of insertions? Show your work for partial credit.

8, 33, 15, 26, 22, 37

b) Open addressing with quadratic probing is used to resolve collisions. What values will be in the hash table after the following sequence of insertions? Show your work for partial credit.

6, 13, 14, 22, 28, 17

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